Regression

Instructor: Chao Huang

Department of Computer Science
Montclair State University

Main sources: CS102 Spring 2020 @Standford

Regression

- Supervised learning
- Model is function from features/predictor values to numeric output
- Example
  - Predictors: mother height, father height, current age
  - Output: height

Regression vs. Classification vs. Clustering

- Regression
  - Supervised learning
  - Numeric output
- Classification
  - Supervised learning
  - Categorical output
  - Example
    - Predictor values: age, gender, income, profession
    - Output: buyer, non-buyer
- Clustering (to be covered next week)
  - Unsupervised
  - Group data into sets of items similar to each other
  - Example
    - Group customers based on purchasing patterns

Regression

- Goal: function f applied to training data should produce values as close as possible in aggregate to actual outputs

Training data

$w_1, x_1, y_1, z_1 \rightarrow o_1$
$w_2, x_2, y_2, z_2 \rightarrow o_2$
$w_3, x_3, y_3, z_3 \rightarrow o_3$
……

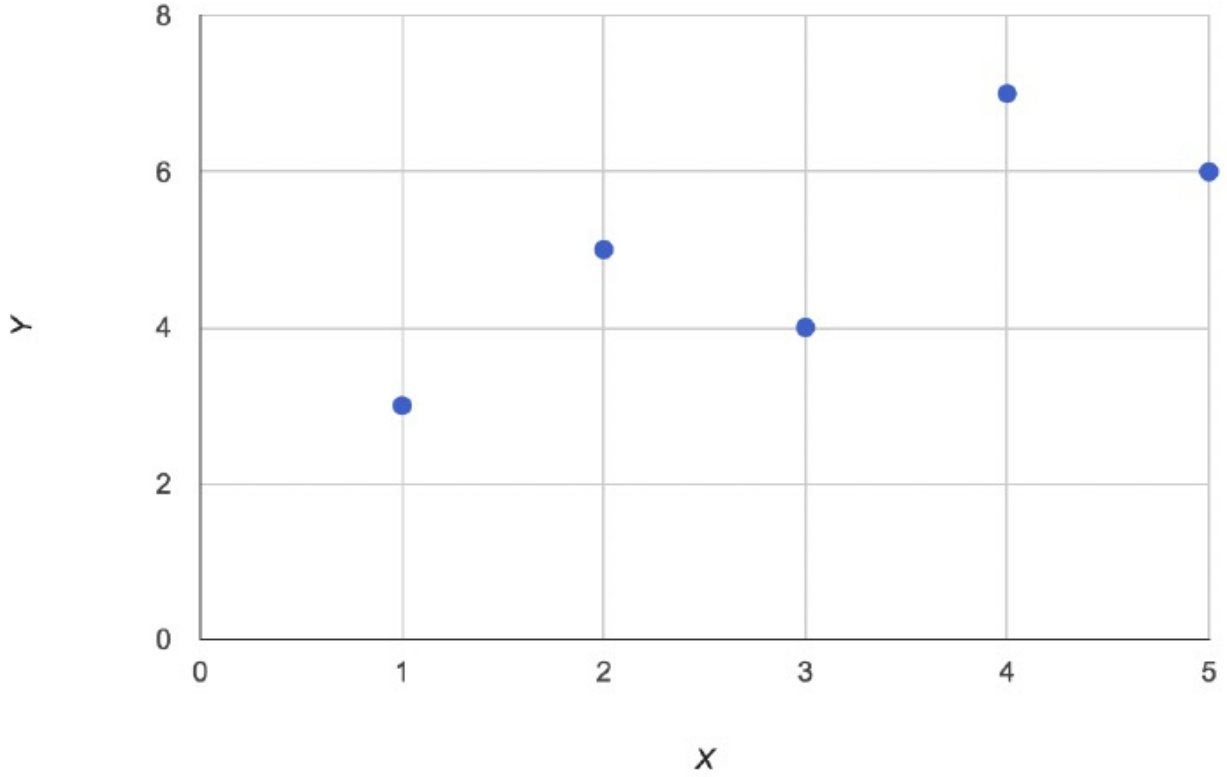
Model

$f(w, x, y, z) = o$

$f(w_1, x_1, y_1, z_1) = o_1'$
$f(w_2, x_2, y_2, z_2) = o_2'$
$f(w_3, x_3, y_3, z_3) = o_3'$

Linear Regression

- Consider
  - One numeric predictor value x
  - One numeric output y
- Data points in two dimensional space

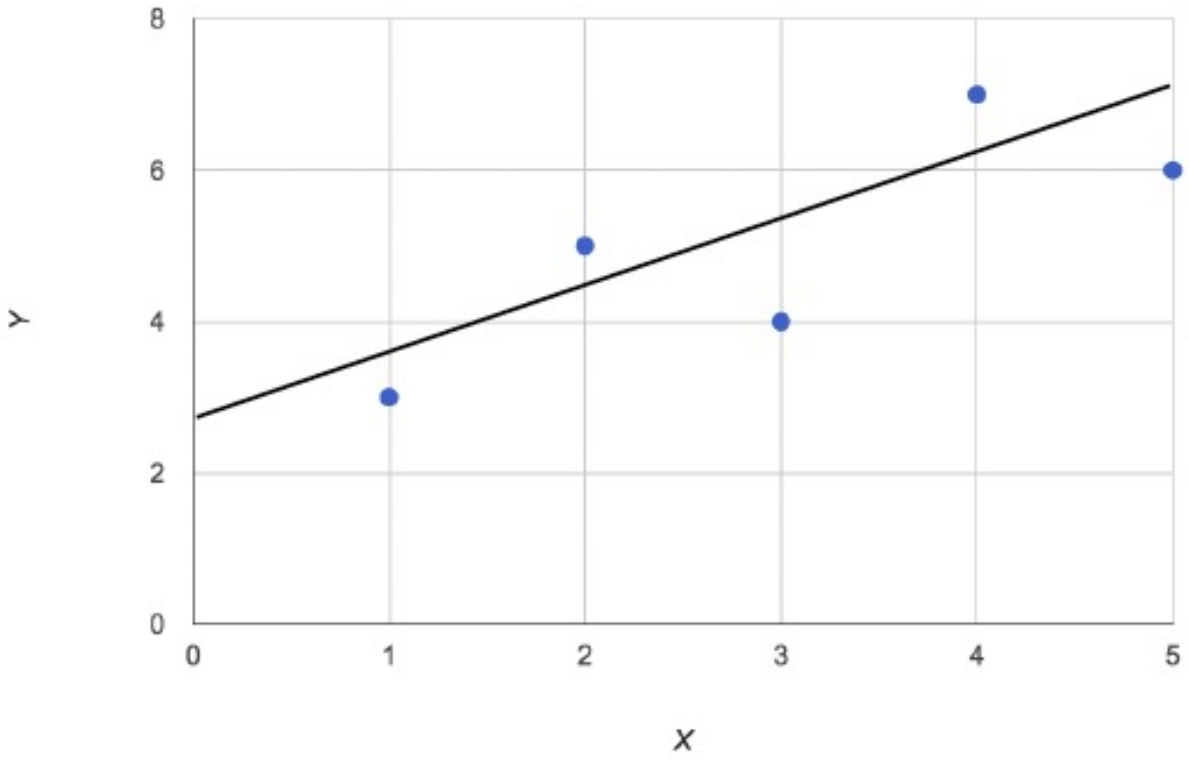


- Goal: find a line that fits the data

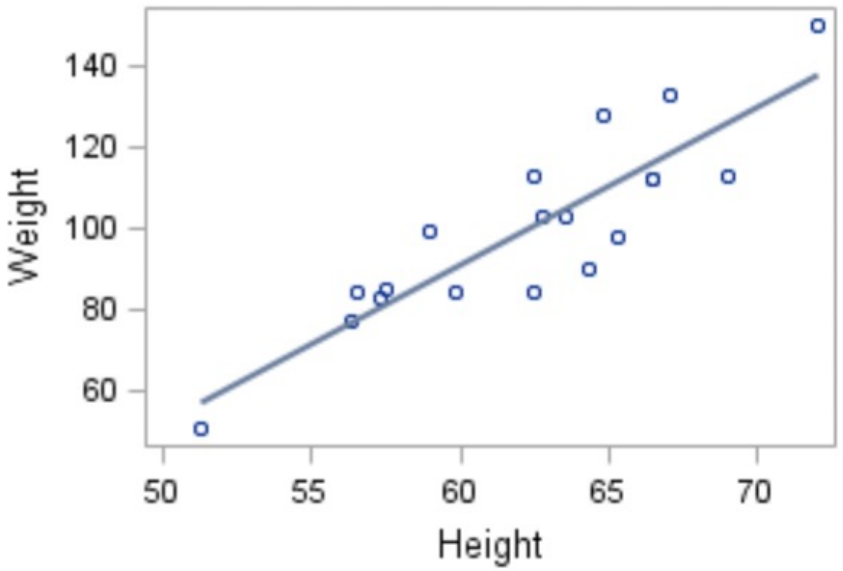
Linear Regression

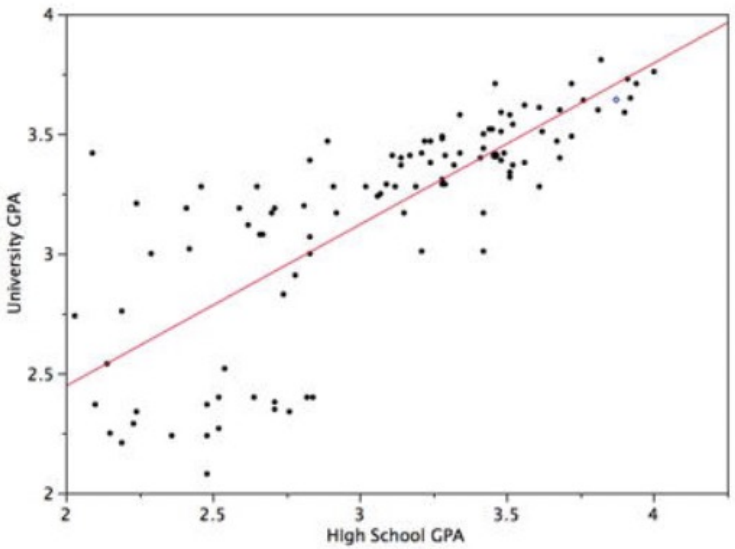
- Functions that are lines: $y = ax + b$

$$y = 0.8x + 2.6$$



Some Examples



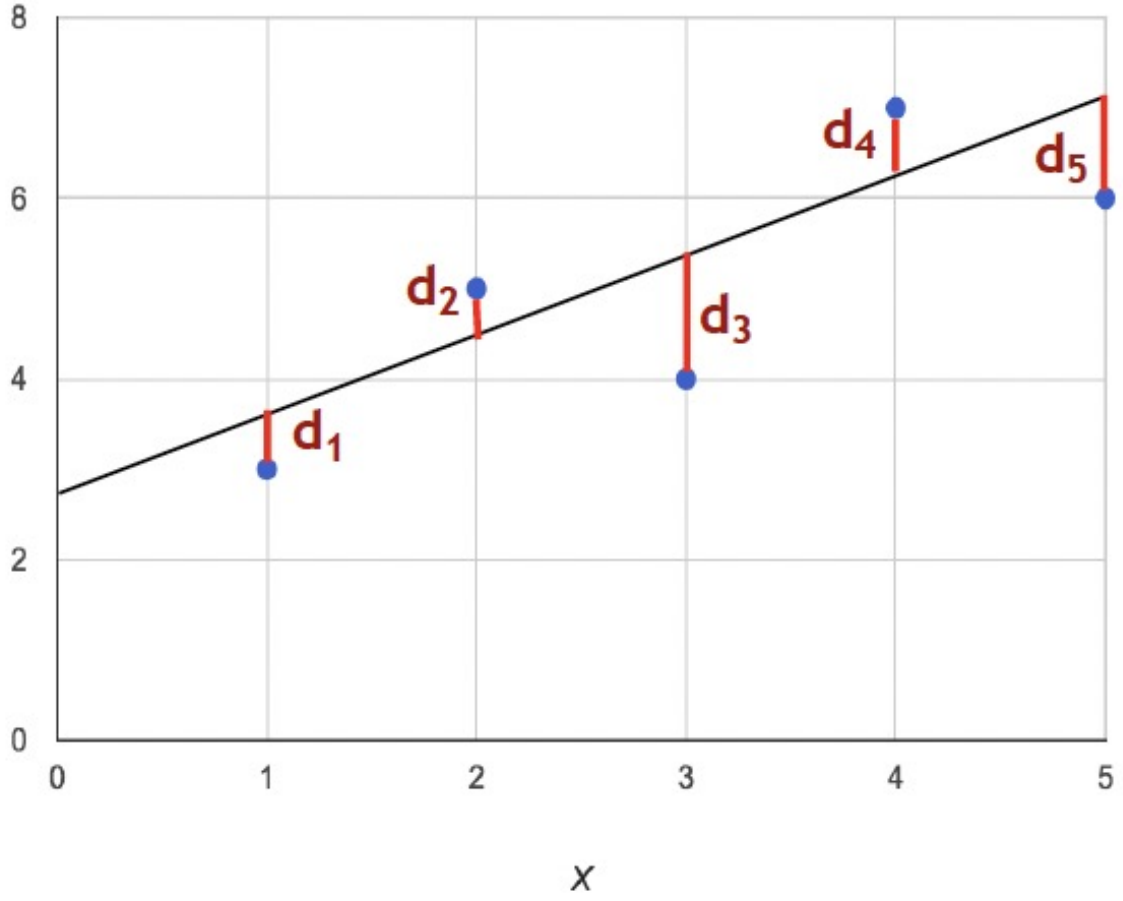


Calculating Linear Regression

Method of least squares

- Given a point and a line, the error for the point is the vertical distance d from the line, and the squared error is d^2
- Given a set of points and a line, the sum of squared error (SSE) is the sum of the squared errors for all points
- Goal: given a set of points, find the line that minimizes SSE

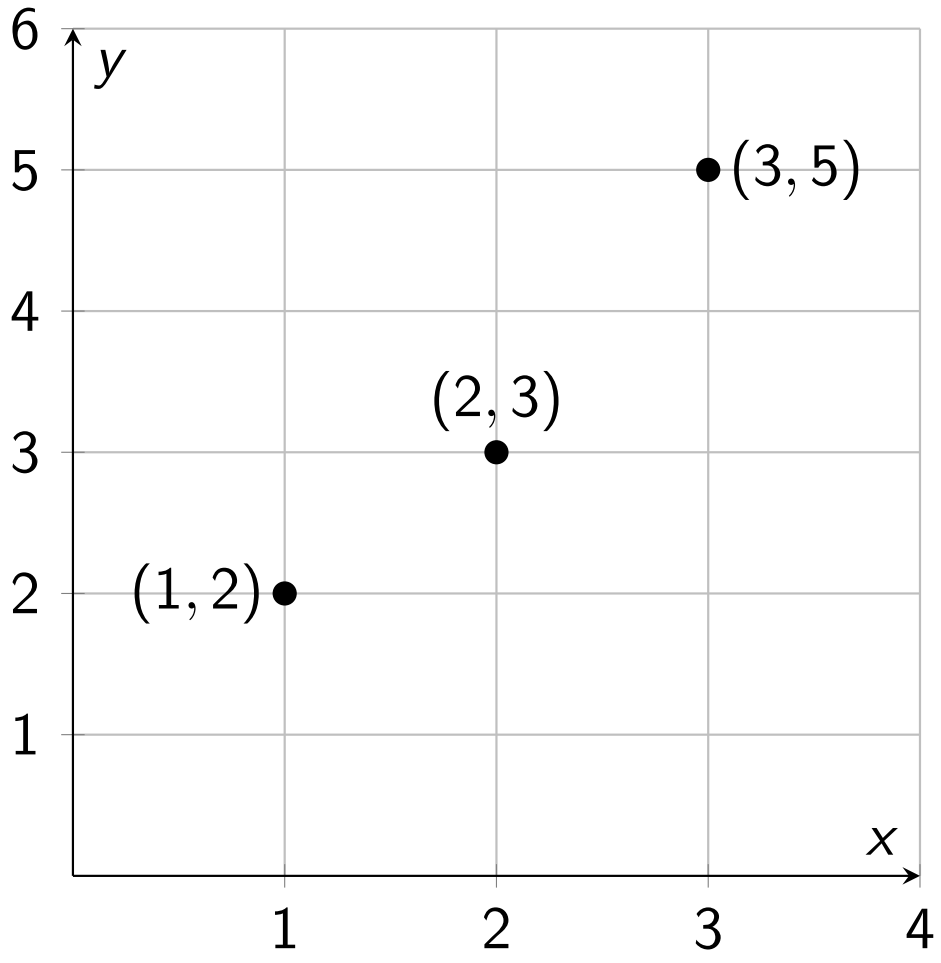
SSE - Example



$$SSE = d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$$

Example

- Consider three points $(1, 2),\ (2, 3),\ (3, 5)$
- Goal: fit a line $y = mx + b$ and calculate the least squared error



Step 1: Determine the Line Equation

- **Formulas for slope m and intercept b:**

$$m = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}, \quad b = \frac{\sum y_i - m\sum x_i}{n}.$$

- Given Points $(x_1, y_1) = (1, 2)$, $(x_2, y_2) = (2, 3)$, $(x_3, y_3) = (3, 5)$
  - Calculate the required sums:

$$\sum x_i = 6, \quad \sum y_i = 10, \quad \sum x_i y_i = 23, \quad \sum x_i^2 = 14$$

Step 1: Determine the Line Equation - Cont'

- Now we calculate the slope m and intercept b:

$$m = \frac{3 \times 23 - 6 \times 10}{3 \times 14 - 6^2} = 1.5$$

$$b = \frac{10 - 1.5 \times 6}{3} = 0.33$$

- The equation of the best-fit line is:

$$y = 1.5x + 0.33$$

  - Q: why is this line "best-fit"?

Step 2: Compute Predicted Values and Squared Errors

- Using the equation $y = 1.5x + 0.33$, we compute the predicted y-values for each point.
  - Predicted values:

$$\hat{y}_1 = 1.83, \quad \hat{y}_2 = 3.33, \quad \hat{y}_3 = 4.83$$

- Now, compute the squared errors:

$$\text{Error}_1 = (2 - 1.83)^2 = 0.0289$$

$$\text{Error}_2 = (3 - 3.33)^2 = 0.1089$$

$$\text{Error}_3 = (5 - 4.83)^2 = 0.0289$$

Step 3: Calculate the Least Squared Error

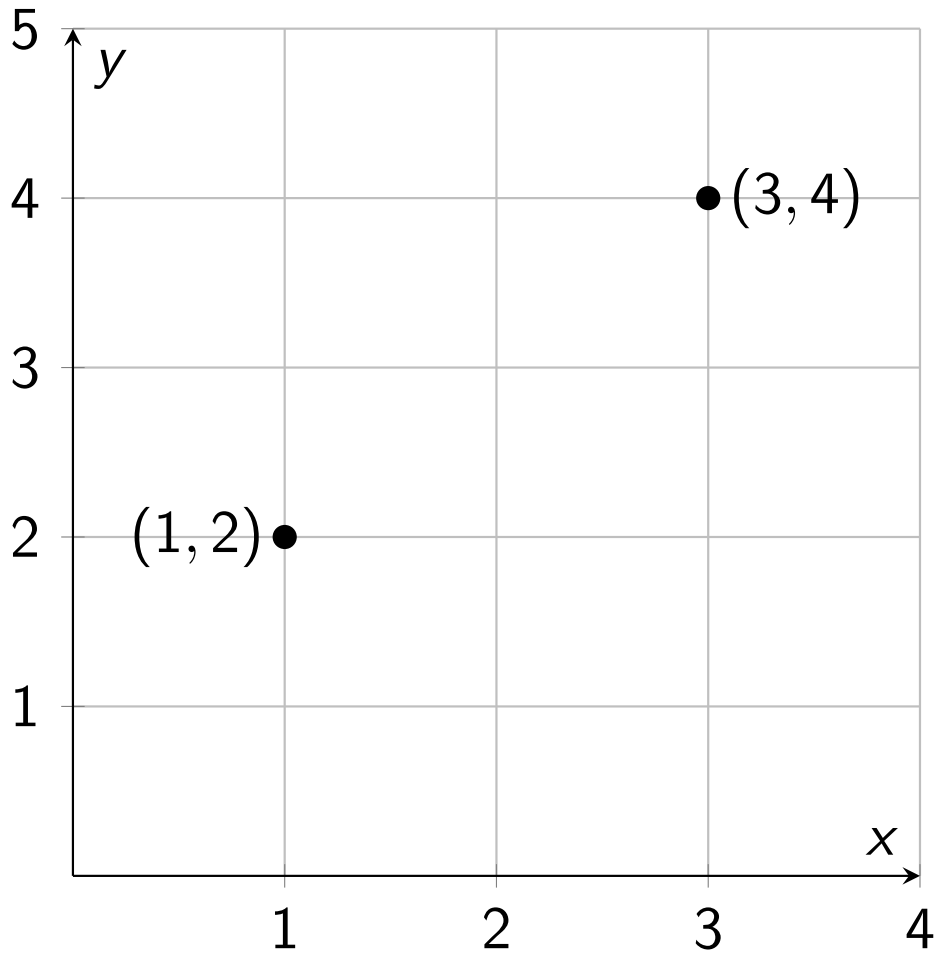
- Finally, sum the squared errors to get the least squared error:

$$\text{LSE} = 0.0289 + 0.1089 + 0.0289 = 0.1667$$

- The least squared error measures how well the line fits the points. In this case, the line doesn't perfectly fit the points, and the error is 0.1667.

Exercise

- Consider two points $(1, 2)$, $(3, 4)$
- Goal: fit a line $y = mx + b$ and calculate the least squared error



Solutions

- Step 1: Determine the Line Equation
- Step 2 - Predicted Values and Squared Errors
- Step 3 - Least Squared Error

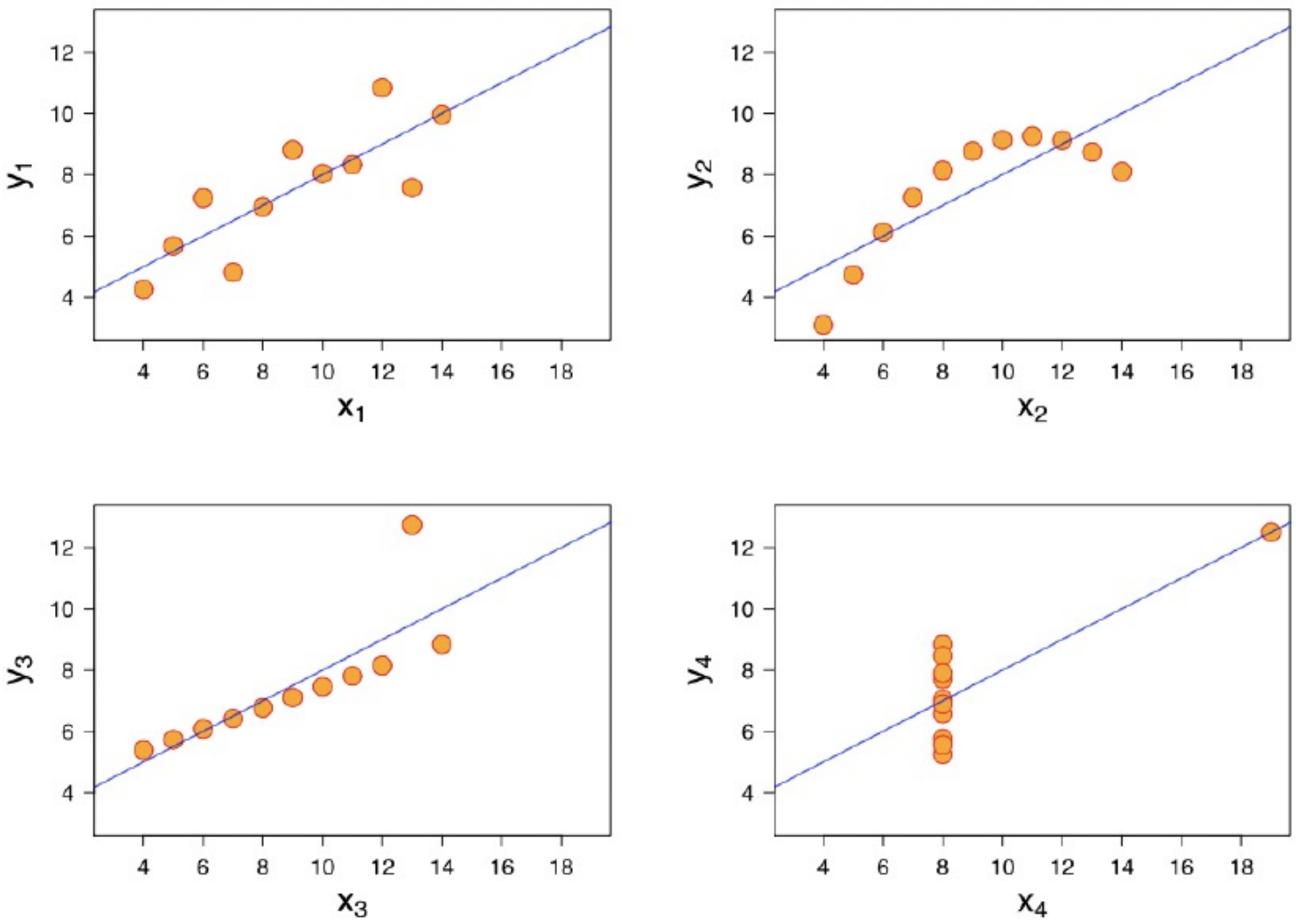
Correlation

- Correlation: values track each other
  - Height and shoe size
  - Grades and SAT scores
- Pearson's Product Moment Correlation (PPMC)
  - "Pearson coefficient", "correlation coefficient"
  - Value between -1 and 1
    - 1: maximum positive correlation
    - 0: no correlation
    - -1: maximum negative correlation
- Coefficient of determination
  - r^2, R^2
  - Measures fit of any line/curve to set of points
  - Usually between 0 and 1
  - For linear regression, $R^2 = Pearson^2$

Correlation Game

- https://www.istics.net/Correlations/

Shortcoming of Linear Regression



- All have same R^2!
- Q: what to do?

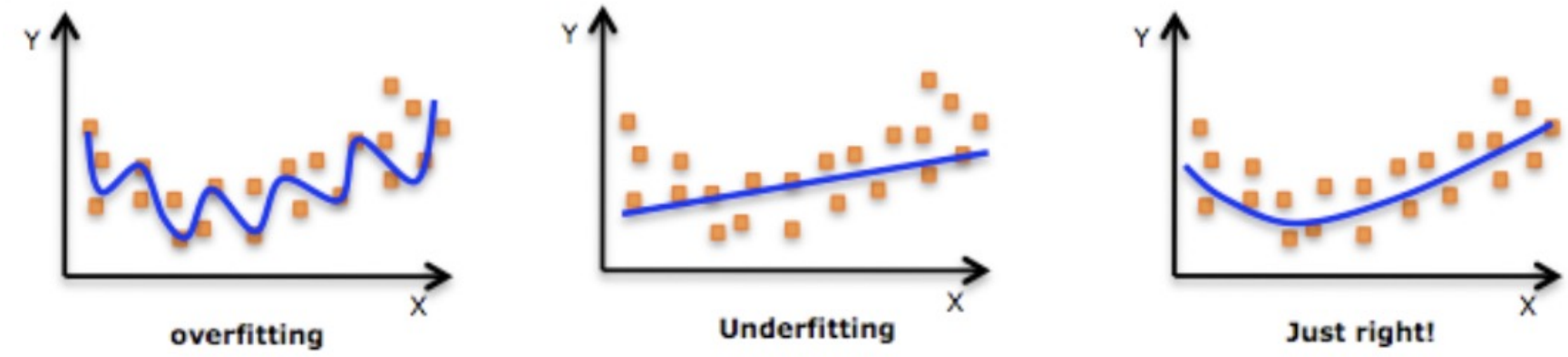
Polynomial Regression

- Given: set of known (x, y) points
- Goal: find function f that "best fits" the known points, i.e., $f(x)$ is close to y

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

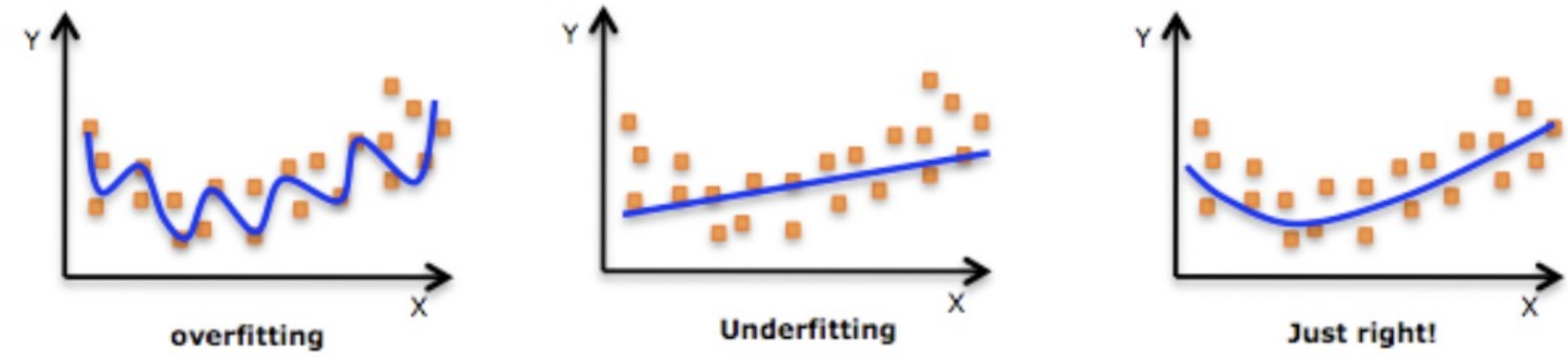
- "Best fit" is still method of least squares
- How to pick n?

Choice of n



- Pick smallest n that fits the data reasonably well

Choice of n



- Pick smallest n that fits the data reasonably well

Regularization

- Traditional linear regression minimizes SSE to find the best-fit line:

$$\min_{w} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

  - w: weights; $\hat{y}_i = f(x_i)$
- However, minimizing the SSE alone can lead to overfitting when the model becomes too complex, fitting the noise in the data instead of generalizing to new data points
- Regularization helps address this issue by penalizing large weights in the model, encouraging simpler models that generalize better.

Ridge Regression

- Ridge Regression addresses overfitting by adding a penalty on the size of the coefficients (weights)
- The objective function for Ridge is:

$$\min_{w} \left(\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} w_j^2 \right)$$

- The term $\lambda \sum_{j=1}^{p} w_j^2$ is called the regularization term, where λ controls the strength of the regularization:
  - If $\lambda = 0$, Ridge reduces to ordinary least squares regression.
  - As λ increases, the model becomes simpler, preventing overfitting.
- The penalty term shrinks the coefficients towards zero but does not eliminate any feature.

Lasso Regression

- Lasso Regression (Least Absolute Shrinkage and Selection Operator) is similar to Ridge, but it uses an $L1$-norm penalty, which can shrink coefficients to zero.
- The objective function for Lasso is:

$$\min_{w} \left(\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} |w_j| \right)$$

- The $L1$-norm penalty encourages sparsity, meaning that some coefficients become exactly zero, performing feature selection.
  - Lasso is particularly useful when we suspect that many features are irrelevant or redundant.
  - Unlike Ridge, Lasso can exclude features from the model entirely.
- The tuning parameter λ controls the trade-off between fitting the data and regularization.

Ridge vs Lasso: Key Differences

- **Ridge Regression:**
  - Uses $L2$-norm penalty $\sum w_j^2$.
  - Shrinks coefficients but does not eliminate them.
  - Suitable when all features are expected to contribute to the model.
- **Lasso Regression:**
  - Uses $L1$-norm penalty $\sum |w_j|$.
  - Can shrink some coefficients to zero, performing feature selection.
  - Useful when there are many irrelevant features.
- Both methods aim to prevent overfitting, but Lasso also helps with feature selection.

Choosing the Regularization Parameter λ

- The regularization strength λ is critical in determining how much to penalize large coefficients.
- A common approach to choose λ is to use cross-validation:
  - Split the data into training and validation sets.
  - Test different values of λ and pick the one that minimizes the validation error.
- A good choice of λ balances bias and variance:
  - Too small λ: Model overfits, high variance.
  - Too large λ: Model underfits, high bias.

Summary

- Linear regression
- Polynomial regression
- Regularization
  - Ridge
  - Lasso